

Architectural Particular Specification

BUILDERS WORK FOR BUILDING SERVICES INSTALLATION

SECTION 1.11

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PART 1 GENERAL

1.01 SCOPE OF WORK

A. The Contractor shall:-

1. Provide and construct all necessary builder's works for the various building services systems and trades for the due installation and works of the respective sub-contractors as detailed in the following sections according to the requirements of respective sub-contractors and according to the builder's work drawings with no adverse comment from the Architect in the way described hereafter.
2. Be responsible for coordinating all relevant sub-contractors for various building services systems on all necessary builder's works, and prepare and submit for the Architect's confirmation of no adverse comment combined builder's works drawings showing consolidated and coordinated details of builder's works of all systems and trades. All builder's work shown in the submitted combined builder's work drawings should be agreed by all relevant sub-contractors before submission to the Architect.
3. Ensure that the said coordination works with sub-contractors and submission of builder's work drawings for the Architect's confirmation of no adverse comment are carried out and completed at appropriate time with reference to the updated construction programme (with reasonable time allowed for Architect's perusal and comment of drawings).
4. Ensure that all builder's works are allowed / provided according to the builder's work drawings approved and agreed by the sub-contractors. The Contractor shall be responsible for any delay or abortive site work / builder's works a result of late and/or poor quality / lack of coordination works and/or submissions.
5. Coordinate with all building services specialists, utility companies, government authorities, Employer's direct contractors, other Main Contractors to ensure the completion of building services installation in full compliance with the drawings and specifications.

B. The Main Contractor shall:-

1. Provide and dismantle temporary scaffolding / working platform for building services installation.
2. Provide permanent hoisting facilities such as lifting eye bolts / hoisting structure I beams as indicated.
3. Construct plant rooms, pipe duct enclosure, etc, with access doors / panels for building services installation.
4. Construct floating slab for designated plant rooms, where applicable.
5. Provide all necessary holes, chases, recesses and openings (temporary or permanent) for the services installation for transportation, hoisting and installation of equipment. Shop drawings shall be submitted for seeking no adverse comment from Architect/Engineer prior to actual making of these openings, chases, etc.
6. Provide concrete curb surround for the openings through floor slab for the passage of services.
7. Provide structural and masonry floors and walls openings; position and cast in sleeves for the above openings; supply, position all cast in box outs for the above openings.

8. Patch and make good of building structure, walls, slabs and ceilings for the services installation.
 9. Seal up the gaps between the sleeves and the services with approved water proof and / or fire proof sealing materials and/or acoustic barrier as appropriate subsequently.
 10. Seal up openings of box outs and the gaps between the sleeves and the structural and masonry floors and walls.
 11. Seal up the gaps between the structural and masonry floors and walls and the fire damper frame.
 12. Provide door louvers or undercut doors in which locations as shown on the louver schedule drawings or the building services design drawings.
 13. Excavate and back fill services trenches.
 14. Provide all concrete water tanks complete with suitable ladders and access manhole.
 15. Provide hoisting beams or hooks cast-in structure inside lift machine rooms, plant rooms for hoisting of service equipment.
 16. Provide fire rated enclosures to the designated areas such as protected lobbies, staircases, etc.
 17. Construct manholes or access hatches with opening frame and cover.
 18. Provide, including supply and install where applicable, the following builder's works in connection with the installation including, but not be limited, to the items listed in the following paragraphs 2 to 10 of this particular specification.
- C. During the Contract period and Defect Liability Period and when required, the Contractor shall remove ceiling tiles and access panels in fixed ceilings as required for testing, commissioning, adjustment and maintenance access for services installation as well as reinstate the ceiling tiles and access panels. Make-good and repair in case damage was caused for purposes stated.
- D. The Contractor shall:-
1. Accurately mark and set out on site the position of all holes and chases, and shall provide supervision on site to ensure that the works are carried out to his requirements. Cutting away and making good shall be kept to a minimum, and this Contractor shall bear the cost of any abortive works, alterations or making good executed by others and caused directly or indirectly by errors or omissions on his part.
 2. Verify dimensions and locations during casting preparation of cast in of sleeves.
 3. Protective barrier (both temporary during works and permanent as necessary) around large openings for safety and to comply with relevant regulations.
 4. Provide a competent person to monitor and coordinated the work with all trades.

1.02 WARRANTY

- A. This part shall be read in conjunction with Specification - Preliminaries for the list of material/ system requiring joint written guarantees.

PART 2 MATERIALS

2.01 MVAC

- A. Concrete sub-bases, plinths, concrete for inertia blocks for chillers, heat pumps, pumps, plate type heat exchangers, water storage tank, solar tube panels, sand filters, air cooled condensers, generators, switchboards, maintenance platforms, system components and all other services equipment.
- B. Structural supports for air conditioning and ventilating equipment.
- C. Weatherproof exterior louvers for air intakes and exhausts.
- D. Louvres shown in Drawings as well as the MVAC installation design drawings.
- E. Aluminium backing sheet of sufficient thickness and stiffeners behind the louvres together with cut-out openings to suit the subsequent installation of fans, air ducts, pipings, etc.
- F. Factory-made proprietary access panels in ceilings, bulkhead, and walls for servicing of equipment. Locations and details of the access panels shall be submitted for Architect's confirmation with no adverse comment before ordering and installation.
- G. Supply and installation of door grilles, weatherproof louvres and decorative wall grilles complete with bird screen.
- H. Openings in floors, walls and roof for passage of ducts, pipes, etc. and subsequent making good with grouting materials to achieve the requisite fire-rating and waterproofing performances. The material shall be submitted to the Architect for confirmation of no adverse comment.
- I. Kerbing (minimum 150mm from finished floor level) and kerb flashing for ducts, pipes, etc. Appropriate shop-drawings illustrating the details of kerb and flashing to be submitted for Architect's confirmation of no adverse comment.
- J. All chases and furring for pipes and ducts.
- K. Positioning, fixing and final firmship of all pipe sleeves as required during construction.
- L. All structural ducts, dog house masonry enclosures, plant rooms etc. in association with the building services installation considered by the Architect or Engineer to be required for the installation during the services coordination stage and may not be shown on the Drawings.
- M. Openings, holes and access panels including necessary supports on suspended ceilings for installation of ceiling diffusers, louvres, air grill and linear diffusers etc.
- N. Provide underflashed penetration with kerbs at the roof slab for exhaust discharges and air intakes.
- O. Bulk-head covering all refrigerant pipes, drain pipes and able to support the installation of A/C split-type DX Fan Coil Units.
- P. Preparing, priming and painting of all exposed surfaces of ductwork, pipework and fixings, except those installed inside Plant Rooms or Pump Rooms. Colour to be approved by Architect.
- Q. Cast-in of sleeves through walls, beams and any structural and non-structural elements.
- R. Permanent hoist beams for the maintenance of mechanical equipment as shown on the drawings.

- S. Shop-drawings, structural details and calculation should be submitted for comment by the structural engineer and to the satisfaction of all relevant government authorities.
- T. Filling of openings outside sleeve with approved waterproof and fire proof sealing materials to acquire the required fire rating or waterproofing properties of the subject wall/floor/roof.
- U. Backfill openings for the passage of ductwork, pipework, cable trays/ducts, busbars, etc. through beams, floors, walls, etc. with approved sealing materials.
- V. All signage with regard to pipe / duct shafts, plant rooms etc. and to comply with all relevant government authorities' requirements.
- W. Fire rated enclosure / ceiling to enclose or sealing system to seal up gaps around the services passing through protected areas such as exit staircases and protected lobbies, or passing through walls with fire resisting period required under the Code of Practice for Fire Resisting Construction 1996.
- X. Supply and installation of stainless steel cover for the condensate drain pipes and refrigerant pipes passing through the external wall on the exterior side.

2.02 UNITARY AIR CONDITIONING UNIT

- A. Access panels in bulkhead for servicing of equipment.
- B. Openings in walls / beams for passage of condensate drain pipes and refrigerant and subsequent making good with grouting materials to achieve the requisite fire rating and water-proofing performances. The material shall be submitted to the Architect for confirmation of no adverse comment.
- C. Positioning, fixing and final firmship of all pipe sleeves as required during construction.
- D. Provision of openings, holes and access panels including necessary supports on bulkhead for installation of indoor unit.
- E. Stainless steel hooks satisfying the fall-arrested safety requirement for anchoring safety belt at each A/C platform.
- F. Bulkhead to conceal all exposed pipework according to the typical details included in the contract drawings.
- G. Fire rated enclosures to enclose or sealing system to seal up gaps around the services that run through fire resisting walls such as walls of kitchens or other walls with fire resisting period required under the Code of Practice for Fire Resisting Construction 1996.
- H. Supply and installation of stainless steel cover for the condensate drain pipes and refrigerant pipes passing through the external wall on the exterior side.

2.03 FIRE SERVICES

- A. Concrete sub-bases, plinths, concrete for inertia block including steel frameworks and reinforcements for all fire pumps.
- B. Design and provide thrust blocks for underground piping.
- C. Supply and installation of all concrete water tanks and positioning and fixing of all puddle flanges, inlet and outlet connectors for concrete water tanks.
- D. Sleeves and puddle flanges through walls and water tank walls meeting the building services installation; verify the dimensions and setting out casting preparation.

- E. All exit signs and signage with regard to pipe shafts, plant and pump rooms, Fire Services inlets (FSI) etc.
- F. Provide all concrete water tanks complete with suitable ladders and access manhole, step irons and covers, and positioning and fixing all puddle flange, inlet and outlet connectors for the concrete water tanks.
- G. Openings in floors, walls and roof for passage of pipes and subsequent making good with grouting materials submitted for the Architect's comment to achieve the requisite fire rating and water-proofing performances.
- H. Kerbing and kerb flashing for ducts, pipes, etc. The kerb should be at 150mm (min.) from finished floor level. Appropriate shop-drawings illustrating the details of kerb and flashing to be submitted for Architect's confirmation of no adverse comment.
- I. Provision of all chases and furring for pipes.
- J. Positioning, fixing of all pipe sleeves and boxing-outs on floor slabs as required during construction.
- K. Access, including temporary openings, for transportation of equipment to plant rooms.
- L. Backfilling and making good of space between wall / floor openings and sleeves with grouting materials to achieve the requisite fire-rating and water-proofing performances. The material shall be submitted to the Architect for confirmation of no adverse comment.
- M. Construction of all FS control room, structural ducts, masonry enclosures, valve pits etc.
- N. Provision of openings, holes, and access panels on suspended ceilings for installation of detectors, sprinklers etc.
- O. Construct reinforced concrete sump pits including cat ladder, tank access and cover.
- P. Preparing, Priming and Painting of exposed pipework and fittings except those installed inside Plant Rooms and Pump Rooms.
- Q. Fire rated enclosures to enclose or sealing system to seal up gaps around the services passing through protected areas such as exit staircases, corridors and protected lobbies, or passing through walls with fire resisting period required under the Code of Practice for Fire Resisting Construction 1996.
- R. Supply and installation of all cabinets, for housing hose reels, hydrants, control valve, and fire services inlet (FSI) etc.
- S. Excavating trenches including backfilling and disposal of surplus excavated materials for underground pipework.
- T. Supply and installation of fire shutters, switches, starters and relay boxes for fire signal.

2.04 ELECTRICAL

- A. Concrete sub-bases and plinths for electrical equipment (e.g. control switch cubicle, transformers).
- B. Openings in floors, walls and roof for passage of busduct, cable trunking etc. and subsequent making good with appropriate grouting materials as approved by the Architect to achieve the requisite fire-rating and water-proofing performances.

- C. Kerbing and kerb flashing for ducts, pipes, etc. The kerb should be at 150mm (min.) from finished floor level. Appropriate shop-drawings illustrating the details of kerb and flashing to be submitted for Architect's confirmation of no adverse comment.
- D. Provision of openings, holes and access panels on suspended ceilings for installation of lighting fittings, CCTV etc.
- E. Preparing, Priming and Painting of exposed trunkings, conduits and cable supports outside plant room.
- F. Fire rated enclosure to enclose or sealing system to seal up gaps around the services passing through protected areas such as exit staircases and corridors and protected lobbies, or passing through walls with fire resisting period required under the Code of Practice for Fire Resisting Construction 1996.
- G. All box for signage with regard to services duct, meter room, switches room, plant rooms etc. power point and fluorescent tubes. Lamp, etc. shall be supplied by electrical contractor.
- H. Excavating trenches including backfilling and disposal of surplus excavated materials for underground cables and cable ducts. The main contractor should also be responsible for the laying of underground lead-in cable ducts, draw-pits, etc. in strict accordance with HEC's requirements.
- I. Construction of manholes, draw-pits, cable joint boxes, metal ducts etc. for underground cable ducting in strict accordance with HEC's requirements.
- J. Construction of Transformer Rooms in strict accordance with HEC's requirements.
- K. Construction of earth pits for electrical earthing system.
- L. Sealing and waterproofing utility company's lead-in ducts.
- M. Provide insulation pad to metal accessories. This shall include but not limited to the following main items:
 - 1. Metal curtain rail
 - 2. Metal towel rail
 - 3. Metal paper-holders
 - 4. Metal door hinge and handle
 - 5. Metal shower holder

2.05 PLUMBING AND DRAINAGE

- A. Supply and installation of complete underground soil and waste, storm water disposal system up to 600mm above finished ground / floor level, and 300mm beyond the perimeter screen wall (towards the basement interior).
- B. Supply and installation of the concrete Work for manholes, grease traps, petrol interceptors, sump pits and trapped gullies, gull pit, velocity reducing chambers valve pits, pump pits, catch pits, sand trap, etc. complete with step irons, covers and associated underground drainage pipe terminated at 600mm above finished floor level etc.
- C. Supply and installation of all concrete water tanks complete with suitable ladders and access manhole, step irons and covers, and positioning and fixing all puddle flange, inlet and outlet connectors for the concrete water tanks.

- D. Concrete plinths, stoolings and concrete for inertia block.
- E. Cutting of masonry opening and making good of the same.
- F. Supply of sanitary fixtures and fittings.
- G. All signage with regard to pipe shafts, plant rooms and pump rooms.
- H. Openings in floors, walls and roof for passage of ducts, pipes, etc. and subsequent making good with grouting materials to achieve the requisite fire-rating and waterproofing performances. The material shall be submitted to the Architect for confirmation of no adverse comment.
- I. Kerbing and kerb flashing for ducts, pipes, etc. The kerb should be at 150mm (min.) from finished floor level. Appropriate shop-drawings illustrating the details of kerb and flashing to be submitted for Architect's confirmation of no adverse comment.
- J. Provision of all chases and furring for pipes.
- K. Positioning, fixing of all pipe sleeves as required during construction.
- L. Forming and constructing all surface water channels, stepped channels and flat channels, together with corresponding covers.
- M. Dog house and channel grating shall be provided for concealing exposed pipework as instructed by the Architect.
- N. Excavating trenches including backfilling and disposal of surplus excavated materials for underground pipework and drainage.
- O. Preparing, priming and painting of all exposed pipework and fittings according to the General Specification on Painting and of colour as per Architect's decision, except those inside Pump Rooms and Plant Rooms, etc.
- P. Fire rated enclosures to enclose or sealing system to seal up gaps around the services passing through protected areas such as exit staircases and protected lobbies, or passing through walls with fire resisting period required under the Code of Practice for Fire Resisting Construction 1996.
- Q. All fittings in grease / petrol interceptor shall be supplied and installed by the Main Contractor.
- R. Supply all (wash basin, shower tray and sink) trap and angle valves.
- S. Underground subsoil drain pipe shall be supplied and installed by the Main Contractor.
- T. Design and provide thrust blocks for underground piping.
- U. Pipe work between cistern and urinals up to and including urinal bottom trap.
- V. Top soil and sub-soil drainage work for planter areas.
- W. Bulkhead to enclose water pipes inside all rooms, except electrical plant rooms.
- X. Supply and installation of concrete bedding / surround for underground plumbing pipework.
- Y. Construction of C.I. grating, brickwork enclosure for planter drains.
- Z. Sleeves and puddle flanges through walls and water tank walls meeting the building services installation; verify the dimensions and setting out casting preparation.

- AA. Construct surface channel with cover grating, detail as shown in Drawings and as approved by Architect.
- BB. Construct reinforced concrete sump pits including cat ladder, tank access and cover.

2.06 LIFT

- A. Construct a properly framed, plumbed (within +25mm), enclosed and ventilated lift shafts with reasonably smooth continuous surface.
- B. Construct lift pits, escalator pits, door sill projecting supports at lift entrance, emergency access doors and lift machine rooms complete with steel cat-ladders and railings, waterproofing facility and notice plates in front of the doors of lift machine rooms.
- C. Set out datum for alignment and levelling of lift door frames.
- D. Provide and dismantle temporary scaffolding inside all lift shafts meeting the lift installation.
- E. Provide temporary guards to all lift well entrances and around floor penetrations whilst erection is in progress. Ensure that these guards are kept in position at all times.
- F. Provision of cat ladders giving easy access to each lift pit but the design and exact location of such cat ladder shall be determined by the lift sub-contractor.
- G. Provision of all necessary work to entrances of lift shaft suitable to accommodate requirements for landing doors or gates, call points, indicators, etc. including filling all gaps between structure and door lining with cement grouting after completion of installation.
- H. Provision for loading materials to each of the roof machine rooms for and strength testing of the lift machine hoist beams to the loading specified by the lift sub-contractor and in accordance with the British Standard CP:114:1957 (or 1969 in Metric Units), including removal of loading materials after test satisfactorily completed.
- I. Provision of temporary guard rails to prevent accidental access to the lift shafts at all points of entry.
- J. Building up to packing behind lift entrance architraves / jambs with concrete / cement mortar and making good finishing around architraves.
- K. Provision and / or cutting all necessary holes, chases and openings and make good after installation of the equipment.
- L. Supplying and fixing all R.C. structural supports, beams, reinforced concrete slabs, etc., necessary for the installation of the machinery, counter weight, buffers, etc.
- M. Providing, maintaining, subsequent altering as required and eventually removing suitable scaffolding to the lift sub-contractor's requirements, and protection of work in progress.
- N. The provision of temporary boxing out of the floor slab and loading support for delivery and installation of lift and escalator equipment; steel hoisting beams and / or 'U' bolts at machine room ceiling for hoisting the lift machinery during installation and for future maintenance.
- O. Construction of metal lining and covers to escalators.

- P. Checking the actual alignment of lift shafts, keeping the maximum tolerance of structural inside faces of lift shaft walls, etc. at any levels not exceeding 6mm in one storey height or in height of 3m and not exceeding 25mm in full building height and carrying out any rectification works if required.
- Q. For lift car interior works, all flooring finishes shall be supplied by the Main Contractor. Installation work shall be carried out by the lift Sub-Contractor.
- R. Separating beam / Hoisting beam / separating screen / hoisting hook shall be supplied and installed by the Main Contractor.
- S. Grouting of door architraves and jambs.
- T. Provision of shaft vents.
- U. Making good / grounding of building walls and floors, all holes, chases, recesses, etc. after fixing.
- V. Provision of temporary central power supply point at each floor level. Extension of power supply and temporary lighting in lift shafts shall be provided by the lift sub-contractor.
- W. Furnish all necessary cement and/or concrete for "Grouting-in" brackets, bolts, etc.
- X. All additional openings, holes through the building structure, partition walls and all concrete bases, supports, ducts, etc. required by the lift sub-contractor beyond those included in the Architectural or Structural drawings will also be carried out by the Main Contractor provided that the contractor shall submit full details of such requirements to the Architect for confirmation of no adverse comment so that due consideration may be given before the Main Contractor commences work in the area concerned.
- Y. Provision of temporary electrical power point for testing and commissioning in cases permanent power supplies are not ready during testing period.
- Z. Provision of scaffolding inside lift shaft.

2.07 GAS

- A. Supply and installation of pipe duct for gas pipe including signage for pipe ducts.
- B. Supply and installation of all pipe sleeves as required during construction.
- C. Construction of balance flue aperture and subsequent making good.
- D. Grouting and sealant of balance flue and the aperture with materials and details and to be submitted to the Architect for confirmation of no adverse comment.

2.08 TELECOMMUNICATION FACILITIES

- A. To construct underground ducts and jointing chambers connecting to the lot boundary as per the requirements and specification laid down by the operators. Jointing chamber frames, covers, PVC ducts and other construction details should strictly follow those provided by the operators. The number of lead-in ducts required should be duly coordinated with the operators by the Main Contractor.
- B. To construct TBE rooms in strict accordance with the requirements and specifications laid down by the operators and as per Architect's drawings.

END OF SECTION